9.1 关系及性质

6. Determine whether the relation R on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

c) $x - y$ is a rational number.

d) $x = 2y$.
e) $xy \geqslant 0$.
g) $x = 1$.
h) $x = 1$ or $y = 1$.
答：c) 自反、对称、非 反对称、传递。
d) 非 自反、非 对称、反对称、非 传递。
e) 自反、对称、非 反对称、非 传递（因 (1,0)and(0,-2) not imply (1,-2)）
g) 非 自反、非 对称、反对称、传递。
h) 非 自反、对称、非 反对称、非 传递（因 (2,1)and(1,2) not imply (2，2)）。

33. Let R be the relation on the set of people consisting of pairs (a, b), where a is a parent of b. Let S be the relation on the set of people consisting of pairs (a, b), where a and b are siblings (brothers or sisters). What are $S \circ R$ and $R \circ S$?

答： $S \circ R = \{(a, b) \mid$ a is a parent of b and b has a sibling$\}$,

$R \circ S = \{(a, b) \mid$ a is an aunt or uncle of b$\}$

9.3 关系的表示

14. Let R_1 and R_2 be relations on a set A represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Find the matrices that represent

a) $R_1 \cup R_2$. **b)** $R_1 \cap R_2$. **c)** $R_2 \circ R_1$.
d) $R_1 \circ R_1$. **e)** $R_1 \oplus R_2$.

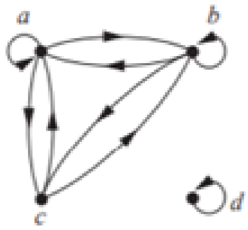
c) The matrix is the Boolean product $\boldsymbol{M}_{R_1} \odot \boldsymbol{M}_{R_2} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

d) The matrix is the Boolean product $\boldsymbol{M}_{R_1} \odot \boldsymbol{M}_{R_1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

答:

32. Determine whether the relations represented by the directed graphs shown in Exercises 26–28 are reflexive, irreflexive, symmetric, antisymmetric, asymmetric, and/or transitive.

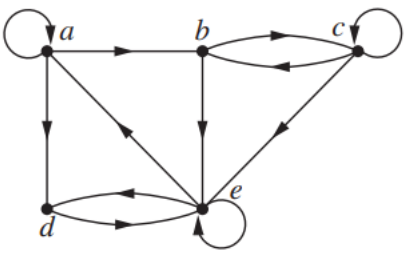
27.



答：仅满足 对称，其它性质都不满足，非 传递((c,a)and(a,c),但没有(c,c))。

9.4 关系闭包运算

17. Find all circuits of length three in the directed graph in Exercise 16.



答：a, a, a, a; a, b, e, a; a, d, e, a; b, c, c, b; b, e, a, b; c, b, c, c;
c, c, b, c; c, c, c, c; d, e, a, d; d, e, e, d; e, a, b, e; e, a, d, e;
e, d, e, e; e, e, d, e; e, e, e, e;

写不全的原因是没有使用 R^3 的算法。

26. Use Algorithm 1 to find the transitive closures of these relations on $\{a, b, c, d, e\}$.

a) $\{(a, c), (b, d), (c, a), (d, b), (e, d)\}$
b) $\{(b, c), (b, e), (c, e), (d, a), (e, b), (e, c)\}$
c) $\{(a, b), (a, c), (a, e), (b, a), (b, c), (c, a), (c, b), (d, a), (e, d)\}$
d) $\{(a, e), (b, a), (b, d), (c, d), (d, a), (d, c), (e, a), (e, b), (e, c), (e, e)\}$

答 :

b) For this and the remaining parts we just exhibit the matrices that arise.

$$\mathbf{A} = \begin{bmatrix} 0&0&0&0&0 \\ 0&0&1&0&1 \\ 0&0&0&0&1 \\ 1&0&0&0&0 \\ 0&1&1&0&0 \end{bmatrix} \qquad \mathbf{A}^{[2]} = \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&1&1&0&0 \\ 0&0&0&0&0 \\ 0&0&1&0&1 \end{bmatrix} \qquad \mathbf{A}^{[3]} = \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&0&1&0&1 \\ 0&0&0&0&0 \\ 0&1&1&0&1 \end{bmatrix}$$

$$\mathbf{A}^{[4]} = \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&1&1&0&1 \\ 0&0&0&0&0 \\ 0&1&1&0&1 \end{bmatrix} = \mathbf{A}^{[5]} \qquad \mathbf{B} = \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&1&1&0&1 \\ 1&0&0&0&0 \\ 0&1&1&0&1 \end{bmatrix}$$

29. Find the smallest relation containing the relation {(1, 2), (1, 4), (3, 3), (4, 1)} that is
a) reflexive and transitive.
b) symmetric and transitive.
c) reflexive, symmetric, and transitive.
答：b)要算 ts(R), c)要算 tsr(R).
b) {(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 4), (3, 3), (4, 1), (4, 2), (4, 4)}
c) {(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 4), (3, 3), (4, 1), (4, 2), (4, 4)}

9.5 等价关系

8. Let R be the relation on the set of all sets of real numbers such that SRT if and only if S and T have the same cardinality. Show that R is an equivalence relation. What are the equivalence classes of the sets {0, 1, 2} and Z?

Proof: show that R is reflexive, symmetric, and transitive.
(1) Every set has the same cardinality as itself, so SRS , R is reflexive.
(2) If SRT , |S| =|T|, then |T| =|S|, TRS, so R is symmetric.
(3) If SRT and TRW, then |S| =|T|, |T| =|W|, so S| =|W|, R is transitive.

16. Let R be the relation on the set of ordered pairs of positive integers such that ((a, b), (c, d)) ∈ R if and only if ad = bc. Show that R is an equivalence relation.

16. This follows from Exercise 9, where f is the function from the set of pairs of positive integers to the set of positive rational numbers that takes (a,b) to a/b, since clearly $ad = bc$ if and only if $a/b = c/d$.

If we want an explicit proof, we can argue as follows. For reflexivity, $((a,b),(a,b)) \in R$ because $a \cdot b = b \cdot a$. If $((a,b),(c,d)) \in R$ then $ad = bc$, which also means that $cb = da$, so $((c,d),(a,b)) \in R$; this tells us that R is symmetric. Finally, if $((a,b),(c,d)) \in R$ and $((c,d),(e,f)) \in R$ then $ad = bc$ and $cf = de$. Multiplying these equations gives $acdf = bcde$, and since all these numbers are nonzero, we have $af = be$, so $((a,b),(e,f)) \in R$; this tells us that R is transitive.

Proof: show that R is reflexive, symmetric, and transitive.

(1) ((a, b), (c, d)) ∈ R if and only if ad = bc.

(a, b), (a, b)) ∈ R because a·b=b·a, R is reflexive.

(2) If ((a, b), (c, d)) ∈ R, then ad = bc, so cb=da, ((c, d), (a, b)) ∈ R, thus R is symmetric.

(3) If ((a, b), (c, d)) ∈ R and ((c, d), (e, f)) ∈ R,

Then ad=bc, cf=de. adcf=bcde, because cd is nonzero, so af=be.

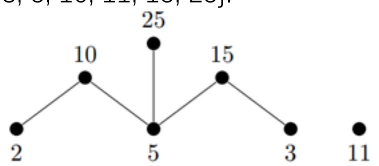
((a, b), (e, f)) ∈ R, thus R is transitive.

9.6 偏序关系

22. Draw the Hasse diagram for divisibility on the set

c) {2, 3, 5, 10, 11, 15, 25}.

答：



35. Answer these questions for the poset ({{1}, {2}, {4},{1, 2}, {1, 4}, {2, 4}, {3, 4}, {1, 3, 4}, {2, 3, 4}}, ⊆).

a) Find the maximal elements.

答：a) {1, 2}, {1, 3, 4}, {2, 3, 4}

44. Determine whether these posets are lattices.

c) (Z, ⩾)

答：this is a lattice because it is a linear order. The least upper bound of two numbers in the list is the smaller number (since here “greater” really means “less”!), and the greatest lower bound is the larger of the two numbers.

补充 9.1 抽象代数与二元运算

24. Let A be a set with n elements. How many binary operations can be defined on A?

Solution：n^{n^2}，不是 2^{n*n}

28. Let $*$ be a binary operation on a set A, and suppose that $*$ satisfies the idempotent, commutative, and associative properties, as discussed in Example 16. Define a relation $\leq$ on A by $a \leq b$ if and only if $b = a * b$. Show that $(A, \leq)$ is a poset and, for all a and b, $\mathrm{LUB}(a, b) = a * b$.

Proof:

1. $\leq$ is reflexive

   Since idempotent, $a = a*a$, $a \leq a$ for all a in A.

2. $\leq$ is antisymmetric

   suppose that $a \leq b$ and $b \leq a$.

   Then, $b = a*b$, and $a = b*a$, because commutative, $a*b = b*a$, so $a = b$.

3. $\leq$ is transitive

   If $a \leq b$ and $b \leq c$, then $c = b*c = (a*b)*c = a*(b*c) = a*c$, so $a \leq c$.

4. show $a*b = a \vee b$, for all a and b in A

(1) $a*b = a*(b*b) = (a*b)*b = b*(a*b)$, so $b \leq a*b$.

In a similar way, $a*b = (a*a)*b = a*(a*b)$, so $a \leq a*b$.

so $a*b$ is a upper bound for a and b.

(2) if $a \leq c$ and $b \leq c$. then $c = a*c$ and $c = b*c$ by definition.

Thus $c = a*(b*c) = (a*b)*c$. so $a*b \leq c$.

This shows that $a*b$ is the leastest upper bound of a and b.

补充 9.2 半群与群

12. Let $S = \{x \mid x \text{ is a real number and } x \neq 0, x \neq -1\}$. Consider the following functions $f_i : S \to S$, $i = 1, 2, \ldots, 6$:

$$f_1(x) = x, \quad f_2(x) = 1 - x, \quad f_3(x) = \frac{1}{x}$$

$$f_4(x) = \frac{1}{1-x}, \quad f_5(x) = 1 - \frac{1}{x}, \quad f_6(x) = \frac{x}{x-1}.$$

Show that $G = \{f_1, f_2, f_3, f_4, f_5, f_6\}$ is a group under the operation of composition. Give the multiplication table of G.

Proof:（1）show composition is closure. 从运算表自然可知.

∘	f1	f2	f3	f4	f5	f6
f1	f1	f2	f3	f4	f5	f6
f2	f2	f1	f5	f6	f3	f4
f3	f3	f4	f1	f2	f6	f5
f4	f4	f3	f6	f5	f1	f2
f5	f5	f6	f2	f1	f4	f3
f6	f6	f5	f4	f3	f2	f1

(2) identity e is f1;
(3) show every element exist reverse.
 $f2^{-1}=f2$, $f3^{-1}=f3$, $f4^{-1}=f5$, $f6^{-1}=f6$.
(4) show associate.
 because ° is associate.

16. Let G be a group with identity e. Show that if $a^2 = e$ for all a in G, then G is Abelian.

Proof: If $a*a=e$, $\forall a\in G$. show commutative.
(1)group propertise $(a*b)^{-1}=b^{-1}*a^{-1}$
because $a*a=e=a*a^{-1}$ so, $b^{-1}*a^{-1}=b*a$,
(2) $(a*b)*(a*b)=e$ => $(a*b)^{-1}=a*b$,
reverse be only one in Group. so $b*a=a*b$.

补充 9.3 子代数与商代数

22. Prove or disprove that the intersection of two subsemigroups of a semigroup $(S, *)$ is a subsemigroup of $(S, *)$

proof: closure;

 $\forall a,b\in S_1\cap S_2$, then $a*b\in S_1$, $a*b\in S_2$, so $a*b\in S_1\cap S_2$.

 if $S_1\cap S_2=\varnothing$, still be subsemigroup.

2. Let $(S, *)$ and $(T, *')$ be monoids. Show that $S \times T$ is also a monoid. Show that the identity of $S \times T$ is (e_S, e_T).

Proof:
(1) *show (S×T,*'') Is a binary operation.*
 Let (s1,t1),(s2,t2) ∈S×T.
 then (s1,t1)''(s2,t2) = (s1*s2, t1*'t2)*
 *so *'' is a binary operation.*
(2) *show *'' Is associative.*
Consider *(s1,t1)*''((s2,t2) *''(s3,t3)) = (s1,t1)*''(s2*s3, t2*'t3)*
 = *(s1*(s2*s3), t1*'(t2*'t3))*
 = ((s1 s2)*s3, (t1*' t2)*'t3)) by (S,*) and (T,*') be associative.*
 *= (s1*s2,t1*'t2)*''(s3, t3)*
 = ((s1,t1)''(s2,t2)) *''(s3,t3))*
 Thus (S×T,'') Is a semigroup.*
(3) *show (S×T,*'') has a identity (es,et).*

Let es is a identity of (S,*), et is a identity of (T,*').

Let (s,t) ∈S×T. then (s,t)''(es,et) = (s*es, t*'et) = (s,t).*

(es,et)''(s,t) = (es*s, et*'t) = (s,t).*

Thus (S×T,'') Is a monoid.*

22. Let G be an Abelian group with identity e, and let $H = \{x \mid x^2 = e\}$. Show that H is a subgroup of G.

Proof:

(1)show H is closed.

Let a,b∈H. (a*b)*(a*b) = a*(b*a)*b = a*(a*b)*b = a^2*b^2=e*e=e.

Thus (a*b) ∈H.

(2)show e∈H.

Because e*e = e, so e∈H.

(3)show if x∈H then x^{-1}∈H.

Let x∈H, then e= e*e =(x* x^{-1}) *(x* x^{-1}) = x* (x^{-1} *x)* x^{-1}

By G is an Abelian group, = x* (x *x^{-1})* x^{-1}

= (x*x) *(x^{-1}* x^{-1}) = e*(x^{-1}* x^{-1}) =(x^{-1}* x^{-1})

Thus x^{-1}∈H.

24. Let G be a group and let $a \in G$. Define $H_a = \{x \mid x \in G$ and $xa = ax\}$. Prove that H_a is a subgroup of G.

Proof: show Ha={x|xa=ax} is a subgroup.

(1)show Ha is closed.

Let x,y∈Ha. (x*y)*a = x*(y*a) = x*(a*y) = (x*a)*y = (a*x)*y = a*(x*y) .

Thus x*y ∈Ha.

(2)show e∈Ha.

Because e*a = a = a*e, so e∈Ha.

(3)show if x∈Ha then x^{-1}∈Ha.

Let x∈Ha, then xa= ax, so a=x^{-1}*(xa) = x^{-1}* (a*x) = (x^{-1}*a)*x

a *x^{-1}=((x^{-1}*a)*x) * x^{-1} = (x^{-1}*a)*(x*x^{-1}) = x^{-1}*a

Thus x^{-1}∈H.

25. Let A_n be the set of all even permutations in S_n. Show that A_n is a subgroup of S_n.

Proof: (1) identity e must be even permutation. So e ∈An.

(2) 证闭合性。设 p1,p2 为两个偶置换，即 p1 和 p2 可对应偶数次转换，p1°p2 必然对应偶数次转换，故 p1°p2∈An.

(3) 逆元，因偶置换 x 只能和偶置换复合得到偶置换 e，An 包含 Sn 所有偶置换，故 x^{-1}∈An.

27. Find all subgroups of the group given in Exercise 17.

答: **27.** $\{f_1\}, \{f_1, f_2, f_3, f_4\}, \{f_1, f_3, f_5, f_6\}, \{f_1, f_3, f_7, f_8\}, \{f_1, f_5\}, \{f_1, f_6\}, \{f_1, f_3\}, \{f_1, f_7\}, \{f_1, f_8\}, D_4$.

补充 9.4 同态定理

28. Let $(S_1, *)$, $(S_2, *')$, and $(S_3, *'')$ be semigroups, and let $f \cdot S_1 \to S_2$ and $g: S_2 \to S_3$ be isomorphisms. Show that $g \circ f: S_1 \to S_3$ is an isomorphism.

proof: $(g°f)(x *_1 y)= g(f(x *_1 y))=g(f(x) *_2 f(y))$

$=g(f(x)) *_3 g(f(y))=(g°f)(x) *_3 (g°f)(y)$.

one-to-one: suppose $(g°f)(a)=(g°f)(b)$, $g(f(a))=g(f(b))$, $f(a)=f(b)$, $a=b$.

onto: $\forall z \in S_3$ 有 $g(y)=z$; $\forall y \in S_2$ 有 $f(x)=y$;

24. Let $A = \{0, 1\}$ and consider the free semigroup A^* generated by A under the operation of catenation. Let N be the semigroup of all nonnegative integers under the operation of ordinary addition.

(a) Verify that the function $f: A^* \to N$, defined by $f(\alpha)$ = the number of digits in α, is a homomorphism.

(b) Let R be the following relation on A^*: $\alpha\ R\ \beta$ if and only if $f(\alpha) = f(\beta)$. Show that R is a congruence relation on A^*.

(c) Show that A^*/R and N are isomorphic.

Proof: Let A={0,1} and consider $(A*, \cdot)$, $(N,+)$.

(1)$f(\alpha)$=length(α), show f:A*->N is homomorphism.

$f(\alpha \cdot \beta)=f(\alpha\beta)$=length($\alpha$)+length($\beta$)=$f(\alpha)+f(\beta)$

(2)R: $f(\alpha)=f(\beta)$,show R is congruence relation.

First show R is equivalence relation.

2. if $f(\alpha)=f(\beta)$ and $f(\chi)=f(\delta)$,then $f(\alpha \cdot \chi) = f(\beta \cdot \delta)$

(3)A*/R and N is isomorphic.

(A*/R,*) : $[\alpha]*[\beta]=[\alpha \cdot \beta]$.

Let $g([\alpha])$=length(α), (1)show g: A*/R->N is homomorphism.

$g([\alpha] *[\beta])=g([\alpha \cdot \beta])$=length($\alpha$)+length($\beta$)=$g([\alpha])+g([\beta])$

(2) onto : $\forall x \in N$, let α=00···0(x factors) ,$g([\alpha])=x$.

(3) one-to-one : suppose $g([\alpha])=g([\beta])$,

length(α)=length(β),so $f(\alpha)=f(\beta)$, then $\alpha R \beta$, $[\alpha]=[\beta]$.

28. **Let G be an Abelian group and n a fixed integer. Prove that the function $f: G \to G$ defined by $f(a) = a^n$, for $a \in G$, is a homomorphism.**

proof: (数学归纳法证)

1.because ab=ba, $ab*ab=aabb=a^2b^2$,

2.$(ab)^n=a^nb^n$, $ab*(ab)^n=ab*a^nb^n=aa^nbb^n=a^{n+1}b^{n+1}$

Hence, $f(ab)=(ab)^n=a^nb^n=f(a)f(b)$.

30. Let G be a group with identity e. Show that the function $f: G \to G$ defined by $f(a) = e$ for all $a \in G$ is a homomorphism.

Proof: Let $a,b \in G$.

$f(a*b)=e = e*e = f(a)*f(b)$

Thus f is a homomorphism.

32. Let G be a group. Show that the function $f: G \to G$ defined by $f(a) = a^{-1}$ is an isomorphism if and only if G is Abelian.

Proof: (1)Suppose f is isomorphism,

$f(xy)=(xy)^{-1}=f(x)f(y)=x^{-1}y^{-1}$

so $xy=((xy)^{-1})^{-1}=(x^{-1}y^{-1})^{-1}=yx$. G is Abelian.

(2)Suppose G is Abelian.

$f(xy)=(xy)^{-1}=x^{-1}y^{-1}=f(x)f(y)$, homomorphism.

onto: $\forall x \in G$, $f(x^{-1})=(x^{-1})^{-1}=x$.

one-to-one: suppose $f(x)=f(y)$, $x^{-1}=y^{-1}$,

$xx^{-1}=e=yy^{-1}$, right cancelation $x=y$.

34. Let $G = \{e, a, a^2, a^3, a^4, a^5\}$ be a group under the operation of $a^i a^j = a^r$, where $i + j \equiv r \pmod 6$. Prove that G and Z_6 are isomorphic.

Proof: Let $f: G \to Z_6$, where $f(a^i) = [i]$. $f(e)=[0]$.

(1) show f is a homomorphism.

Let $a^i, a^j \in G$, $f(a^i * a^j) = f(a^r) = [r] = [I] +_6 [j] = f(a^i) +_6 f(a^j)$

Thus f is a homomorphism.

(2) show f is onto.

Let $[x] \in Z_6$, then $f(a^x)=[x]$.

(3)show f is one-to-one.

Let $f(a^x)= f(a^y) =[x]$. so $x \equiv y \pmod 6$,

But 0~5 is distinct in (mod 6), so $x=y$, $a^x= a^y$

Thus f is one-to-one.

4. Let G_1 and G_2 be groups. Show that the function $f: G_1 \times G_2 \to G_1$ defined by $f(a, b) = a$, for $a \in G_1$ and $b \in G_2$, is a homomorphism.

Proof: Let (a,b) and (c,d) $\in G_1 \times G_2$.

$f((a,b)*''(c,d))=f(a*c,b*'d)=a*c=f(a,b)*f(c,d)$

Thus f is homomorphism.

10. Let $G = S_3$. Determine all the left cosets of $H = \{f_1, g_1\}$ in G.

Solution:

If $a \in H$, then $aH = H$. Thus $f_1 H = g_1 H = H$.

$f_2 H = \{f_2, g_3\}$

$f_3 H = \{f_3, g_2\}$

$g_2 H = \{g_2, f_3\} = f_3 H$

$g_3 H = \{g_3, f_2\} = f_2 H$

The distinct left cosets of H in S_3 are H, $f_2 H$, and $f_3 H$.

18. Let N be a subgroup of a group G, and let $a \in G$. Define

$$a^{-1}Na = \{a^{-1}na \mid n \in N\}.$$

Prove that N is a normal subgroup of G if and only if $a^{-1}Na = N$ for all $a \in G$.

Proof: (1)If N is normal subgroup,

then $\forall a \in G$, aN=Na, and $e \in N$.

so $a^{-1}Na = Na^{-1}a = Ne = N$.

(2)If $a^{-1}Na = N$ for all $a \in G$,

let $n \in N$, $a^{-1}na = n'$ for some $n' \in N$.

$a \cdot a^{-1}na = a \cdot n'$, $na = an' \in aN$, Then $Na \subseteq aN$.

Similarly, let $n \in N$, $n = a^{-1}n'a$, $an = a a^{-1}n'a = n'a$, Then $aN \subseteq Na$.

Hence, aN=Na, N is normal subgroup.

20. Find all the normal subgroups of S_3.

Solution: (1)find all subgroups.

{f1},{f1,g1},{f1,g2},{f1,g3},{f1,f2,f3},{f1,f2,f3,g1,g2,g3}

(2) select normal subgroups.

{f1}, {f1,f2,f3}, {f1,f2,f3,g1,g2,g3}.

28. Let G be an Abelian group and N a subgroup of G. Prove that G/N is an Abelian group.

Proof:

(1)As G be an Abelian group,

then $\forall a \in G$, $\forall h \in N$, ah=ha; so aN=Na,

then N is a normal subgroup.

(2) G/N is $=\{aN \mid a \in G\}$, N = eN =[e].

(aN)(bN) = [a]*[b] = abN, by Theroem 3 corollary 1,

as ab=ba, abN = baN = [b]*[a]=(bN)(aN).

Hence, G/N is an Abelian group.

30. Let H and N be subgroups of the group G. Prove that if N is a normal subgroup of G, then $H \cap N$ is a normal subgroup of H.

Proof: Prove that if N is a normal subgroup of G, then $H \cap N$ is a normal subgroup of H.

(1)show $H \cap N$ is a subset of H;

$\forall x \in H \cap N$, $x \in H$, $H \cap N$ is a subset of H.

(2)show $H \cap N$ is a closure;

$\forall x,y\in H \cap N$, $x\in H$ and $y\in H$, $x*y\in H$; $x\in N$ and $y\in N$, $x*y\in N$; so $x*y\in H \cap N$.

(3)Since H and N are subgroup of G, $e\in N$ and $e\in H$,

so $e\in H \cap N$.

(4)$\forall x\in H \cap N$, $x \in H$, $x^{-1} \in H$, and $x \in N$, $x^{-1} \in N$;

so $x^{-1} \in H \cap N$.

(5) $\forall x\in H$, $x(H\cap N) = \{xn|\ n\in H\cap N\}$, let any $n_1\in H\cap N$,

because aN=Na, $x*n_1=n_2*x$ for some $n_2 \in N$,

$n_2= x*n_1*x^{-1}$,

since $x^{-1} \in H$, $n_1\in H$, then $n_2 \in H$.

$n_2\in H\cap N$,

Thus $x(H\cap N) = (H\cap N)x$.

Hence, $H \cap N$ is a normal subgroup of H.

31. Let $f : G \to G'$ be a group homomorphism. Prove that f is one to one if and only if $\ker(f) = \{e\}$.

Proof: (1)show if ker(f)=e,then f is one to one.

Suppose ker(f)=e. Let $g1,g2\in G$, and f(g1)=f(g2).

Then $f(g1g2^{-1})=f(g1)f(g2^{-1})=f(g1)(f(g2)^{-1})=f(g1)(f(g1)^{-1})=e$.

Hence $g1g2^{-1}\in \ker(f)$, thus $g1g2^{-1}=e$ and g1=g2.

So f is one-to-one.

(2)show if f is one to one, then ker(f)=e.

Suppose f: G->G' is one-to-one. Let $x\in \ker(f)$.

Then f(x)= e' = f(e). Thus x=e and ker(f)={e}.

补充 11.1 群编码

18. Show that the (3, 7) encoding function $e: B^3 \to B^7$ defined by

$$
\begin{array}{ll}
e(000) = 0000000 & e(100) = 1000101 \\
e(001) = 0010110 & e(101) = 1010011 \\
e(010) = 0101000 & e(110) = 1101101 \\
e(011) = 0111110 & e(111) = 1111011
\end{array}
$$

is a group code.

Proof:

(1) e of B^7 is e(000). (2) $\forall a\in e(B^3)$, $a\oplus a=e$, reverse exist.

(3) list multiplication table, $(e(B^3),\oplus)$ is closure.

⊕	e(000)	e(001)	e(010)	e(011)	e(100)	e(101)	e(110)	e(111)
e(000)	e(000)	e(001)	e(010)	e(011)	e(100)	e(101)	e(110)	e(111)
e(001)	e(001)	e(000)	e(011)	e(010)	e(101)	e(100)	e(111)	e(110)
e(010)	e(010)	e(011)	e(000)	e(001)	e(110)	e(111)	e(100)	e(101)
e(011)	e(011)	e(010)	e(001)	e(000)	e(111)	e(110)	e(101)	e(100)
e(100)	e(100)	e(101)	e(110)	e(111)	e(000)	e(001)	e(010)	e(011)
e(101)	e(101)	e(100)	e(111)	e(110)	e(001)	e(000)	e(011)	e(010)
e(110)	e(110)	e(111)	e(100)	e(101)	e(010)	e(011)	e(000)	e(001)
e(111)	e(111)	e(110)	e(101)	e(100)	e(011)	e(010)	e(001)	e(000)

27. Prove the Theoroem 4.

Theorem 4 Let **D** and **E** be $m \times p$ Boolean matrices, and let **F** be a $p \times n$ Boolean matrix. Then

$$(\mathbf{D} \oplus \mathbf{E}) * \mathbf{F} = (\mathbf{D} * \mathbf{F}) \oplus (\mathbf{E} * \mathbf{F}).$$

That is, a distributive property holds for ⊕ and ∗. ●

Proof: (D⊕E)∗F = (D(I,j) ⊕E(I,j))∗F

= ((D(I,1) ⊕E(I,1))∧F1j ⊕ (D(I,2) ⊕E(I,2) ∧F2j ⊕ ⋯ ⊕ (D(I,p) ⊕E(I,p) ∧Fpj)

因为异或和与运算满足分配律,

= [(D(I,1) ∧F1j) ⊕ (E(I,1))∧F1j) ⊕(D(I,2) ∧F2j) ⊕ (E(I,2))∧F2j) ⊕ ⋯ ⊕(D(I,p) ∧Fpj) ⊕ (E(I,p))∧Fpj)

=(D∗F)⊕(E∗F) 因异或运算满足交换律

补充 11.2 群译码

13. Consider the (3, 5) group encoding function $e: B^3 \to B^5$ defined by

$e(000) = 00000$ $e(100) = 10011$
$e(001) = 00110$ $e(101) = 10101$
$e(010) = 01001$ $e(110) = 11010$
$e(011) = 01111$ $e(111) = 11100.$

Decode the following words relative to a maximum likelihood decoding function.

(a) 11001 (b) 01010 (c) 00111

Solution:

(a) min{δ(11001,e(b))}=δ(11001,01001)=1,

d(11001)=010.

(b) min{δ(01010,e(b))}= δ(01010,11010) or δ(01010,01001) =1,

d(01010)=110 or 010.

(c) min{δ(00111,e(b))}=δ(00111,01111) or δ(00111,00110) =1, d(00111)=011 or 001.

18. $\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

m=3,r=3,n=6. $e(B^m)=\begin{bmatrix} 000 \\ 001 \\ 010 \\ 011 \\ 100 \\ 101 \\ 110 \\ 111 \end{bmatrix} * \begin{bmatrix} 100100 \\ 010110 \\ 001011 \end{bmatrix} = \begin{bmatrix} 000000 \\ 001011 \\ 010110 \\ 011101 \\ 100100 \\ 101111 \\ 110010 \\ 111001 \end{bmatrix}$

000000	001011	010110	011101	100100	101111	110010	111001
000001	001010	010111	011100	100101	101110	110011	111000
000010	001001	010100	010011	100110	101101	110000	111011
000100	001111	010010	011001	100000	101011	110110	111101
001000	000011	011110	010101	101100	100111	111010	110001
010000	011011	000110	001101	110100	111111	100010	101001
000101	001110	010011	011000	100001	101010	110111	111100
001100	000111	011010	010001	101000	100011	111110	110101

coser leader={000000,000001,000010,000100,001000,010000, 000101 or 011000 or 100001, 001100 or 010001 or 101000}

23. Let

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function associated with e_H.

(a) 10100 (b) 01101 (c) 11011

Solution:

m=2, r=3,n=5.

$$B^2 * H = \begin{bmatrix} 00 \\ 01 \\ 10 \\ 11 \end{bmatrix} * \begin{bmatrix} 10011 \\ 01101 \end{bmatrix} = \begin{bmatrix} 00000 \\ 01101 \\ 10011 \\ 11110 \end{bmatrix}$$

Decoding table:

00000	01101	10011	11110
00001	01100	10010	11111
00010	01111	10001	11100
00100	01001	10111	11010
01000	00101	11011	10110
10000	11101	00011	01110
00110	01011	10101	11000
01010	00111	11001	10100

Syndrome list:

$$\varepsilon * H = \begin{bmatrix} 00000 \\ 00001 \\ 00010 \\ 00100 \\ 01000 \\ 10000 \\ 00110 \\ 01010 \end{bmatrix} * \begin{bmatrix} 011 \\ 101 \\ 100 \\ 010 \\ 001 \end{bmatrix} = \begin{bmatrix} 000 \\ 001 \\ 010 \\ 100 \\ 101 \\ 011 \\ 110 \\ 111 \end{bmatrix}$$

(a) (10100)*H=(111), => ε = 01010
xt⊕ε=11110, => 11.
(b) (01101)*H=(000),=> ε = 00000
xt⊕ε=01101, => 01.
(c) (11011)*H=(101),=> ε = 01000
xt⊕ε=10011, =>10.